

PULSE WIDTH MODULATION OF LIGHT EMITTING DIODES IN A SPREAD SPECTRUM CONTROLLED SYSTEM AND ANALYSIS IN OCTAVE

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Nível: Masters

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Goals



To discuss how varying PWM duty cycles affect LEDs output parameters



To analyze spread spectrum control and PWM performance using Octave simulation.



To give insight into the use of PMW as a reliable and efficient way to control LED dimming and flickering .

Introduction



LED DIMMING



Material

Software: Octave, Microsoft Applications

Hardware: Computer system

Methods

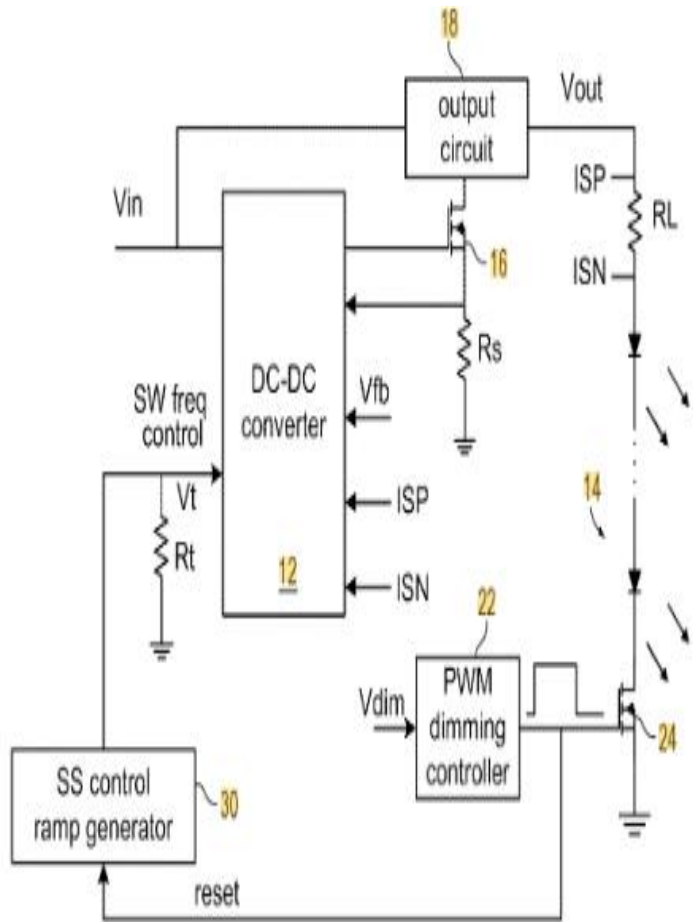
A. Design

B. Simulation

C. Analysis

D. Validation

• Design



The Structure of a Spread Spectrum Controlled PWM LED driver system is shown in Fig.1 above. The operational Principle is as follows;

1. DC-DC Converter (Block 12)

Converts an input voltage V_{in} to a regulated output voltage or current suitable for driving LEDs.

Switching frequency is modulated by the SSC circuit to spread EMI. Feedback signals ISP and ISN are used to regulate the output current via sensing resistor R_s .

2. Output Circuit (Block 18)

Manages LED load R_L .

Supplies regulated voltage V_{out} to the LEDs and helps isolate or protect the converter output.

3. LED String (Block 14)

Represents multiple LEDs connected in series.

The current through this string is the critical parameter that is being regulated.

4. Switching Frequency Control

Controlled by resistor R_t and voltage V_t , which determine the switching frequency of the DC-DC converter.

These parameters can be dynamically altered by the SSC Control Ramp Generator to introduce frequency modulation.

5. SSC Control Ramp Generator (Block 30)

Generates a slowly varying signal (e.g., triangular or sine wave) that modulates the switching frequency.

This reduces the spectral concentration of EMI and synchronizes frequency variation across PWM cycles to prevent visible flicker.

A reset signal aligns the SSC ramp to the start of each PWM pulse.

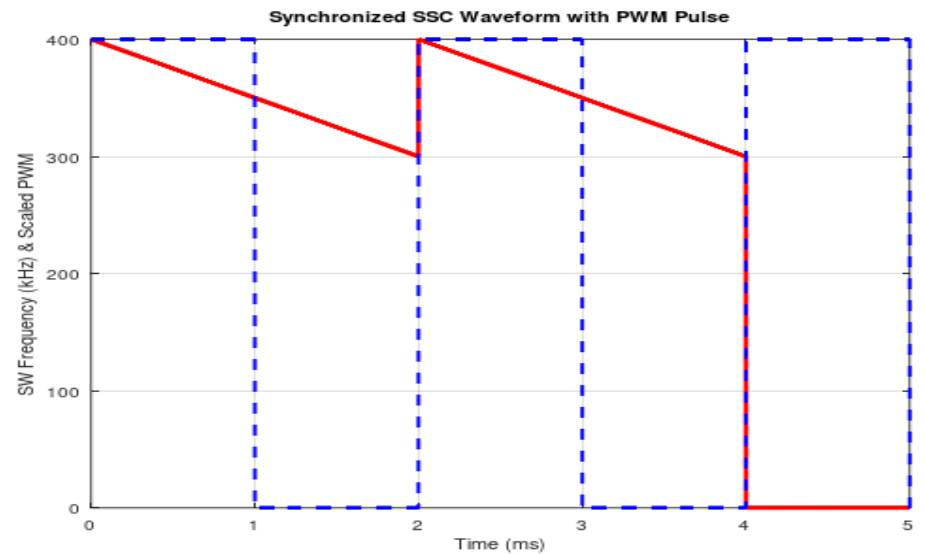
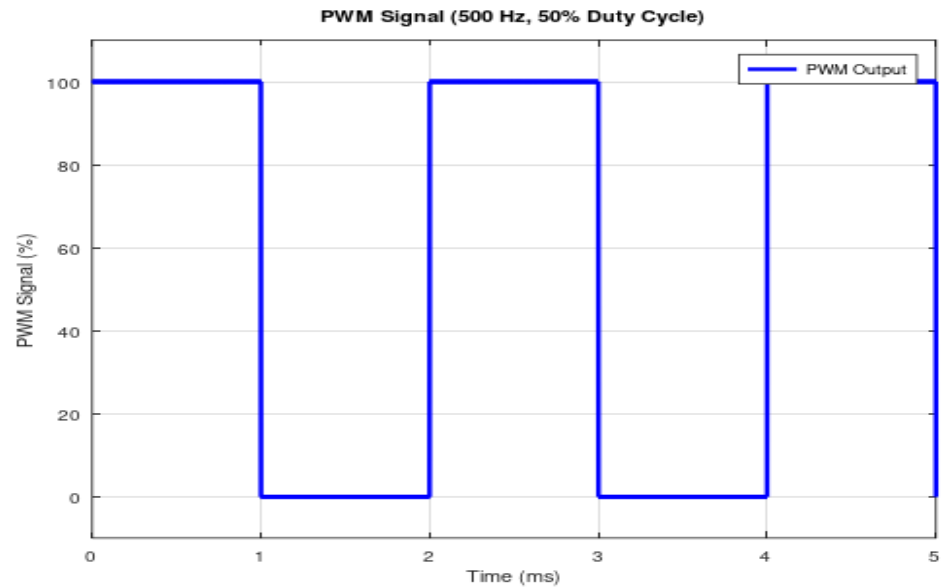
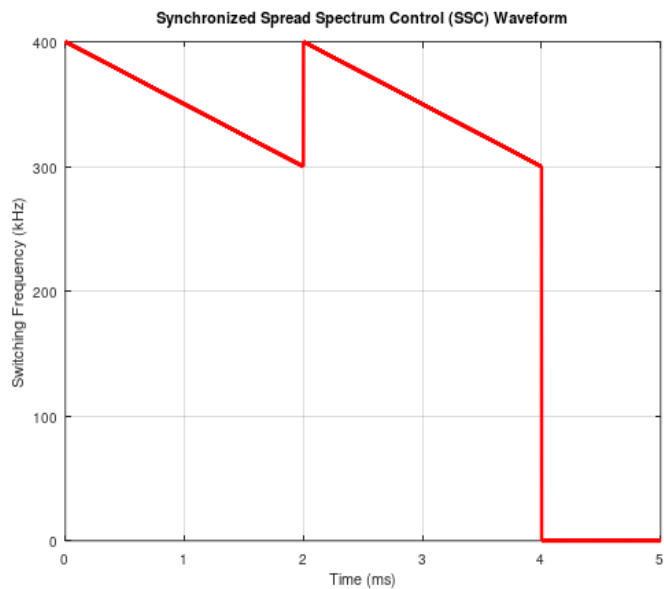
6. PWM Dimming Controller (Block 22)

Receives a dimming control voltage V_{dim} and outputs a PWM signal.

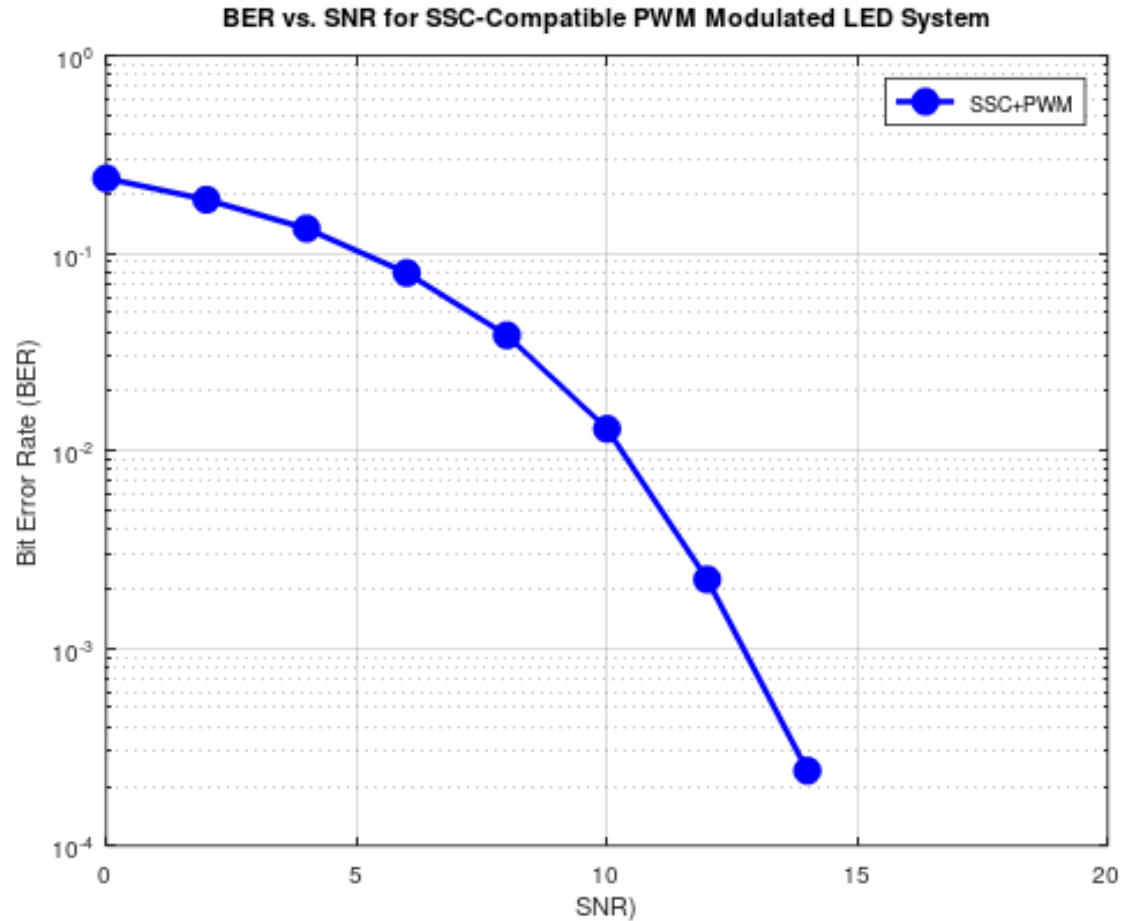
This PWM signal drives the switch (24) in series with the LED string.

Controls the on/off duration of LED current, adjusting brightness and minimizing flicker.

Simulation Results



Simulation Results



Discussion

The findings demonstrate the efficacy of the Pulse Width Modulation method implemented in Octave for LEDs systems. Synchronization ensures the SSC ramp resets with each PWM pulse, producing identical current waveforms cycle-to-cycle, eliminating perceptible flicker

Bit Error Rate BER :

The BER performance was measured at various SNR values ranging from 0 to 20db, results showed that raising SNR resulted in a considerable reduction in BER, which shows that the system is resistant to noise.

At SNR 13db, BER was at around 10^{-3} , showing the system efficiency at high SNR

Signal to Noise Ratio (SNR):

The system performed reliably even at low SNR levels, with a substantial improvement in BER as SNR climbed.

Conclusion

This work shows that employing spread spectrum control synchronized with the PWM dimming signal, the switching frequency of the LED driver converter is modulated in a way that maintains waveform consistency. This results in a perceptible flicker-free LED operation, suitable for both low and high brightness levels.

Future work might include optimizing the system for real-time applications and investigating its performance in more complicated situations.

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